

- The figures in the right margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | Marks |
|---|-------|
| 1. (a) Define fluid, real fluid, ideal fluid, laminar flow and turbulent flow with examples. | 05 |
| (b) Derive equation of continuity for incompressible fluid in vector form and in Cartesian coordinate system. | 05 |
| 2. (a) Define streamline and pathline. Show that the velocity potential satisfies the Laplace equation for incompressible fluid. | 05 |
| (b) If gas flows from a vessel through a small orifice from a region where the pressure is p_1 to a region where the pressure is p_2 . Prove that the rate of flow of mass is $c_2 \rho_2 w_2 \sqrt{\frac{2}{\gamma-1} \left[\left(\frac{p_2}{p_1} \right)^{\frac{1}{\gamma-1}} - 1 \right]}$, where w_2 is the area of the vena contracta and c_2 is the speed of sound in gas, $c_2^2 = \frac{\gamma p}{\rho_2}$, $p = k \rho^\gamma$ and ρ_2 being the density of the vena contracta. | 05 |
| 3. (a) Derive Euler's dynamical equation of motion in vector form. | 05 |
| (b) Show that $u = -\frac{2xyz}{(x^2+y^2)^2}$, $v = \frac{(x^2-y^2)z}{(x^2+y^2)^2}$, $w = \frac{y}{x^2+y^2}$ are the velocity components of a possible motion. Is this motion is irrotational? | 05 |
| 4. (a) Find the centre of pressure of triangular area, depth of whose vertices α, β, γ from the free surface | 05 |
| (b) Find the condition that the components of x, y, z per unit mass acting on a fluid particle may keep in the fluid in equilibrium. | 05 |

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | |
|---|----|
| 5. (a) Define two dimensional source and sink. Obtain the velocity and complex potential due to sources. | 05 |
| (b) Two sources of equal strength are situated respectively at the points $(\pm a, 0)$ in an unbounded fluid. Show that at any point on the circle $x^2 + y^2 = a^2$, the fluid velocity is parallel to the axis of y . | 05 |
| 6. (a) State and prove Blasius theorem. | 05 |
| (b) Find the complex potential due to doublet. | 05 |
| 7. (a) State and prove circle theorem. | 05 |
| (b) Determine the complex potential for a streaming motion past a circular cylinder. | 05 |
| 8. (a) Show that the vorticity vector $\bar{\Omega}$ of an incompressible fluid moving under no external forces satisfies the differential equation $\frac{d\bar{\Omega}}{dt} = (\bar{\Omega} \cdot \nabla) \bar{q} + \nu \nabla^2 \bar{\Omega}$, where ν is the kinematic coefficient of viscosity. | 05 |
| (b) If E is the internal energy of a gas, S is the entropy, Show that $P = -\frac{\partial E}{\partial V}$, $T = \frac{\partial E}{\partial S}$, where all symbols have their usual meanings. | 05 |

***** THE END *****

BSc
MSc
20

31-08-21

Khulna University, Khulna
Mathematics Discipline
4th Year, 1st Term Examination; Session: 2019-2020
Course No: **Math-4105 (old)**; Title of the Course: **Hydrodynamics**
Full Marks: 60, Time: 03 Hours

- The figures in the right margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

- | | | | |
|----|-----|---|----|
| 1. | (a) | Define stream line. Prove that the stream function is constant along the stream line. Explain the rotational and irrotational motion with geometrical interpretation. | 05 |
| | (b) | Derive continuity equation in vector form with geometrical interpretation. | 05 |
| 2. | (a) | Define stagnation point with example. Deduce $U_r = -\frac{1}{r} \frac{\partial \psi}{\partial \theta}$ and $U_\theta = \frac{\partial \psi}{\partial r}$. | 05 |
| | (b) | Show that $U = 2cxy, V = c(a^2 + x^2 - y^2)$ are the velocity components of a possible fluid motion. Find the stream function and sketch the stream line. | 05 |
| 3. | (a) | Derive the Euler's equation on motion in the form $\frac{d\vec{q}}{dt} = F - \frac{1}{\rho} \nabla p$. | 05 |
| | (b) | State and prove Kelvin's minimum energy theorem. | 05 |
| 4. | (a) | Define complex potential and velocity of a fluid motion. State and prove circle theorem. | 05 |
| | (b) | Explain the motion represented by $W = \frac{1}{2} U \frac{a^3}{z^2}$ and show that the stream lines are lemniscates. | 05 |

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | | | |
|----|-----|--|----|
| 5. | (a) | Derive the equation of source and sink in a uniform strength. Also find for equal strength. | 05 |
| | (b) | Find the source and sink of uniform stream parallel to the x-axis. | 05 |
| 6. | (a) | Define vortex line. Deduce the equation of stationary vortex filament in presence of cylinder. | 05 |
| | (b) | If there are sources at $(a,0)$ and $(-a,0)$, sink at $(0,a)$ and $(0,-a)$ with strength m , then show that the stream line through the four points is a circle. | 05 |
| 7. | (a) | Define vortex tube and vortex filament. Find the equation of the vortex lines. | 05 |
| | (b) | Find the equation of complex plane potential of a two dimensional vortex. | 05 |
| 8. | (a) | Find the centre of pressure of a triangle area, depth of whose vertices are α, β, γ from the free surface. | 05 |
| | (b) | A regular hexagon is immersed in a liquid with one side in the surfaces. Show that the depth of its centre of pressure is to that of its centre of gravity as 23:18. | 05 |

***** THE END *****

Mathematics Discipline

Khulna University

4th year, Term-I, Examination; Session: 2020-2021

Course No: Math-4105, Course Title: Hydrodynamics

Full Marks: 60, Time: 03 Hours, Credit Hour: 03

- Figures in the right margin indicate full marks. Questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. a) Define path lines and streamlines. Derive the equation of path lines and streamlines in Cartesian co-ordinate system. 05
- b) State and prove Bernoulli's theorem for incompressible fluid. 05
2. a) Derive the equation of continuity for compressible and incompressible fluid in vector and Cartesian co-ordinate system. 05
- b) Derive the acceleration equation of a fluid particle in Cartesian co-ordinate system. 05
3. a) Derive Euler's equation of motion in vector form as well as Cartesian co-ordinate system. 05
- b) Derive the energy equation of motion of the following form: 05

$$\frac{d}{dt}(T + V + J) = \int_s p(qs) ds$$

↓
n

where all symbols have their usual meaning.
4. a) State and prove Kelvin's minimum energy theorem. 05
- b) Show that $\frac{x^2}{a^2} \tan^2 t + \frac{y^2}{b^2} \cot^2 t = 1$ is a possible form for the boundary surface of a liquid and find an expression for the normal velocity. 05

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. a) State and prove the theorem of Blasius. 05
- b) Determine the motion represented by $w = \frac{1}{2} U \frac{a^3}{z^2}$ and show that the streamlines are lemniscates. 05
6. a) State and prove circle theorem. 05
- b) Define circular vortex and vortex line. Determine the equation of the vortex lines. 05
7. a) Determine the complex potential for a doublet. 05
- b) A source and a sink of equal strength are placed at the point $(\pm \frac{a}{2}, 0)$ respectively, with a fixed circular boundary $|z| = a$, show that the streamlines are given by

$$\left(r^2 - \frac{a^2}{4}\right)(r^2 - 4a^2) - 4a^2 y^2 = ky(r^2 - a^2)$$

where k is a constant. 05
8. a) Find the centre of pressure of a triangular area depth of whose vertices are α, β, γ form the free surface. 05
- b) A plane triangular area is immersed in a liquid of uniform density with its plane vertical, one side horizontal and the opposite corner downwards. Its vertical altitude is 'h' and its horizontal side is at a depth 'h' below the effective surface. Show that its centre of pressure is at the depth $\frac{11}{8} h$ below the surface. 05

***** The End *****

Khulna University, Khulna
Mathematics Discipline
4th Year, Term-I Examination, Session: 2021-2022
Course No: Math-4105
Full Title of Course: Hydrodynamics
Full Marks: 60, Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | | Marks |
|----|---|--------------|
| 1. | <p>(a) Define stream line and path lines. Show that $u = \frac{-2xyz}{(x^2 + y^2)^2}$, $v = \frac{(x^2 - y^2)z}{(x^2 + y^2)^2}$, $w = \frac{y}{x^2 + y^2}$ are the velocity components of the possible fluid motion. Is this motion irrotational?</p> <p>(b) Derive Euler equation of motion. Hence deduce $\frac{\partial \bar{q}}{\partial t} + \bar{\nabla} \left(\frac{q^2}{2} \right) + \bar{\Omega} \times \bar{q} = \bar{F} - \frac{1}{\rho} \bar{\nabla} p$, where $\bar{\Omega} = \text{curl } \bar{q}$</p> | 05

05 |
| 2. | <p>(a) Find the condition that the surface $F(x, y, z, t) = 0$ may be a boundary surface.</p> <p>(b) Determine whether the motion specified by $\bar{q} = \frac{Ax\hat{j} - y\hat{i}}{x^2 + y^2}$, ($A = \text{constant}$) is a possible motion for an incompressible fluid. If so, determine the equation of stream lines. Also show that the motion is of potential kind.</p> | 05
05 |
| 3. | <p>(a) Deduce the Bernoulli's theorem for compressible (barotropic flow)</p> <p>(b) A stream in a horizontal pipe, after passing a contraction in the pipe at which its sectional area is A, is delivered at atmospheric pressure at a place where the sectional area is B. Show that if a side tube connected with the pipe at the former place, water will be sucked up through it into the pipe from a reservoir a depth $\frac{s^2}{2g} \left(\frac{1}{A^2} - \frac{1}{B^2} \right)$ below the pipe; where s is the delivery per second.</p> | 05
05 |
| 4. | <p>(a) Derive continuity equation for compressible fluid with geometrical interpretation. Define stagnation points. Show that the velocity potential $\phi = \frac{1}{2} \log \frac{(x+a)^2 + y^2}{(x-a)^2 + y^2}$ gives a possible motion and determine the form of the streamlines.</p> <p>(b) In a two-dimensional motion show that a streamline cuts itself at a point of zero velocity and that the two branches are at right angles when the motion irrotational. Sketch the streamline which passes through the stagnation point of the motion given by $\psi = U \left(y - a \tan^{-1} \frac{y}{x} \right)$.</p> | 05

05 |

Section-B

Marks

There are **FOUR** questions in this section. Answer any **THREE** questions

5. Define source, sink and doublet with geometrical interpretation. Two sources of strength m (each) are placed at the points $(a, 0)$ and $(-a, 0)$ and a sink of strength $2m$ is placed at the origin. Show that the streamlines are curves $(x^2 + y^2)^2 = a^2(x^2 - y^2 + \lambda xy)$, where λ is the variable parameter. Show also that the fluid speed at any point is $\frac{2ma^2}{r_1 r_2 r_3}$ where r_1, r_2, r_3 are respectively the distance of the points from the sources and sink. 10

6. (a) Define stagnation point. In two-dimensional flow of a liquid, the complex potential is $W = U(z + a^2/z) + ik \log \frac{z}{a}$ where U, k, a are real and positive. Show that (i) the $|z| = a$ is a stream line. (ii) there are in general two stagnation points (iii) the circulation about the circle $|z| = a$ is $2\pi k$. 05
 (b) Find the center of pressure of a triangular are depth of whose vertices are α, β, γ from the free surface. 05

7. (a) A fine circular tube of uniform bore is half filled with equal volumes of four liquids which do not mix and whose densities are as 1:4:8:7 and is held with in a plane vertical. Show that the diameter joining the free surfaces makes an angle $\tan^{-1} 2$ with the vertical. 05
 (b) A triangle is immersed in a homogeneous liquid, show that the sum of the pressure at the vertices is three times the pressure at the centre of the triangle. 05

8. (a) Define Centre of pressure. Show that the depth of centre of pressure of a plane area immersed in a fluid under the action of gravity is greater than the depth of C.G. and the difference of their depth. Prove that the position of the centre of a plane area is independent of inclination of the area to the vertical. 05
 (b) A triangle, of base a and altitude h , is placed in water with it's plane vertical and the side a horizontal and at a depth k below the surface of the water; show that the depth of its center of pressure is $\frac{6k^2 - 4kh + h^2}{2(3k - h)}$. 05

*** END ***

Khulna University, Khulna
Mathematics Discipline
 4th Year, Special Term Examination; Session: 2022-2023
 Course No: **Math-4105**
 Full Title of Course: **Hydrodynamics**
 Full Marks: 60, Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use a separate sheet for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | | Marks |
|----|--|--------------|
| 1. | (a) What do you mean by stream line and path line? Derive a relation between material derivative, local derivative, and convective derivative. | 05 |
| | (b) Define viscosity with geometrical interpretation. State and prove Bernoulli's theorem for incompressible fluid. | 05 |
| 2. | (a) Establish the general form of Euler's dynamical equation of motion of a liquid. | 05 |
| | (b) Derive the energy equation of a fluid motion under conservative forces in the form $\frac{d}{dt}(T + V + J) = \int p q \, ds$, where the symbols have their usual meaning. | 05 |
| 3. | (a) Define complex potential and velocity of a fluid motion. State and prove the circle theorem. | 05 |
| | (b) Show that $u = \frac{-2xyz}{(x^2 + y^2)^2}$, $v = \frac{(x^2 - y^2)z}{(x^2 + y^2)^2}$, $w = \frac{y}{x^2 + y^2}$ are the velocity components of the possible fluid motion. Is this motion irrotational? | 05 |
| 4. | (a) Prove that the streamlines and the potential curves are orthogonal and also show that ϕ and ψ are harmonic, where a two-dimensional irrotational motion is considered. | 05 |
| | (b) Show that if the body force is conservative and the flow is of potential kind, Euler's equation of motion for unsteady flow can be expressed as $\nabla(-\frac{\partial \phi}{\partial t} + \Omega + \int \frac{dp}{\rho} + \frac{q^2}{2}) = \underline{q} \times \underline{\xi}$; where $\underline{\xi}$ is the velocity vector. Hence from it obtain the Bernoulli's equation for steady and irrotational flow. | 05 |

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | | |
|----|---|-----------|
| 5. | (a) Define sink and doublet. Deduce the complex potential for a source. | 05 |
| | (b) Discuss fully the motion arising from the combination of source of strength m at $(a, 0)$ and an equal sink at $(-a, 0)$ in a uniform stream U parallel to the real axis and flowing from a source to sink. | 05 |
| 6. | (a) State and prove Blasius theorem | 05 |
| | (b) Describe the image system due to a source placed in front of a circle. | 05 |
| 7. | (a) Define circulation. Show that circulation can be considered as the measure of velocity potential around a closed circuit. | 05 |
| | (b) State and prove circle theorem. | 05 |
| 8. | (a) State and prove Kelvin's minimum energy theorem. | 05 |
| | (b) A regular hexagon is immersed in a liquid with one side on the surface. Show that the depth of its centre of pressure is to that of its centre of gravity as 23:18. | 05 |

***** THE END *****



Khulna University, Khulna
Mathematics Discipline

4th Year, Term-I, Examination-2024, Session: 2022-2023
Course No: **Math-4105**, Full Title of Course: **Hydrodynamics**
Full Marks: 60, Time: 03 Hours

- The figures in the right margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

- Define stream tube, stream filament, rotational and irrotational flow. Distinguish between stream line and path line. **05**
 - Show that $u = \frac{(3x^2 - r^2)}{r^3}$, $v = \frac{3xy}{r^3}$, $w = \frac{3xz}{r^3}$ are velocity components of possible incompressible fluid flow. Also find the stream line of the flow. **05**
- Find the velocity potential(ϕ) and stream function(ψ) from the complex potential function $w = az^2$ and show them graphically. Also find the fluid velocity **05**
 - Show that the velocity potential $\phi = \frac{1}{2}a(x^2 + y^2 - 2z^2)$ satisfies the Laplacian equation. Also determine the streamlines. **05**
- State Bernoulli's equation. Find the adiabatic expansion for the Bernoulli's equation. Hence show that $\int \frac{dp}{\rho} + \frac{1}{2}q^2 + \Omega - \frac{\partial \phi}{\partial t} = F(t)$, where the symbols have their usual meanings. **05**
 - Show that $\frac{x^2}{a^2 k^2 t^4} + kt^2 \left(\frac{y^2}{b^2} + \frac{z^2}{c^2} \right) = 1$ is a possible boundary surface of a liquid at time t . **05**
- Derive the momentum equation for incompressible fluid flow of the form $\frac{\partial q}{\partial t} + \frac{1}{2} \nabla q^2 - q \times \text{curl} q = F - \frac{1}{\rho} \nabla p$ **05**
 - A stream is rushing from a boiler through a conical pipe, the diameters of the end of which are D and d ; if V and v be the corresponding velocities of the steam and if the motion be supposed to be that of divergence from the vertex of the cone, then prove that $\frac{v}{V} = \frac{D^2}{d^2} \times e^{\frac{v^2 - V^2}{2k}}$, where k is the pressure divided by density and supposed constant. **05**

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

- Define source, sink and doublet with geometrical interpretation. A source and a sink of equal strength are placed at the points $(a,0)$ and $(-a,0)$ respectively, with in a fixed circular boundary $|z|=2a$, show that the streamlines are given by curves $16a^2 y^2 + \lambda y(r^2 - 4a^2) = (r^2 - 16a^2)(r^2 - a^2)$, where λ is a constant. **10**
- State and prove circle theorem. **05**
 - Describe the image system due to a source placed in front of a circle. **05**
- Discuss complex potential and magnitude of complex potential. Show that the stream function and velocity potential satisfy the Cauchy-Riemann equation. **05**
 - Describe the stagnation point. If $w = 2z + 3iz^2$, find the velocity q and stagnation point. **05**
- Explain the vortex and strength of vortex. What is circular vortex? Find the complex potential of a vortex tube. **04**
 - If the velocity of an incompressible fluid at the point (x, y, z) is given by $\left(\frac{3xz}{r^5}, \frac{3yz}{r^5}, \frac{3z^2 - r^2}{r^5} \right)$, then prove that the liquid motion is possible and that the velocity potential is $\frac{\cos \theta}{r^2}$. Also determine the streamlines. Show that the motion is irrotational. **06**

*****The End*****



- The figures in the right margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

- (a) Define stream tube, stream filament, rotational and irrotational flow. Distinguish between stream line and path line. **05**

(b) Show that $u = \frac{(3x^2 - r^2)}{r^5}$, $v = \frac{3xy}{r^5}$, $w = \frac{3xz}{r^5}$ are velocity components of possible incompressible fluid flow. Also find the stream line of the flow. **05**
- (a) Find the velocity potential(ϕ) and stream function(ψ) from the complex potential function $w = az^2$ and show them graphically. Also find the fluid velocity **05**

(b) Show that the velocity potential $\phi = \frac{1}{2}a(x^2 + y^2 - 2z^2)$ satisfies the Laplacian equation. **05**

Also determine the streamlines.
- (a) State Bernoulli's equation. Find the adiabatic expansion for the Bernoulli's equation. Hence show that $\int \frac{dp}{\rho} + \frac{1}{2}q^2 + \Omega - \frac{\partial \phi}{\partial t} = F(t)$, where the symbols have their usual meanings. **05**

(b) Show that $\frac{x^2}{a^2 k^2 t^4} + kt^2 \left(\frac{y^2}{b^2} + \frac{z^2}{c^2} \right) = 1$ is a possible boundary surface of a liquid at time t . **05**
- (a) Derive the momentum equation for incompressible fluid flow of the form $\frac{\partial q}{\partial t} + \frac{1}{2} \nabla q^2 - q \times \text{curl} q = F - \frac{1}{\rho} \nabla p$ **05**

(b) A stream is rushing from a boiler through a conical pipe, the diameters of the end of which are D and d ; if V and v be the corresponding velocities of the steam and if the motion be supposed to be that of divergence from the vertex of the cone, then prove that $\frac{v}{V} = \frac{D^2}{d^2} \times e^{\frac{v^2 - V^2}{2k}}$, where k is the pressure divided by density and supposed constant. **05**

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

- Define source, sink and doublet with geometrical interpretation. A source and a sink of equal strength are placed at the points $(a,0)$ and $(-a,0)$ respectively, with in a fixed circular boundary $|z|=2a$, show that the streamlines are given by curves $16a^2 y^2 + \lambda y(r^2 - 4a^2) = (r^2 - 16a^2)(r^2 - a^2)$, where λ is a constant. **10**
- (a) State and prove circle theorem. **05**

(b) Describe the image system due to a source placed in front of a circle. **05**
- (a) Discuss complex potential and magnitude of complex potential. Show that the stream function and velocity potential satisfy the Cauchy-Riemann equation. **05**

(b) Describe the stagnation point. If $w = 2z + 3iz^2$, find the velocity q and stagnation point. **05**
- (a) Explain the vortex and strength of vortex. What is circular vortex? Find the complex potential of a vortex tube. **04**

(b) If the velocity of an incompressible fluid at the point (x, y, z) is given by $\left(\frac{3xz}{r^5}, \frac{3yz}{r^5}, \frac{3z^2 - r^2}{r^5} \right)$, then prove that the liquid motion is possible and that the velocity potential is $\frac{\cos \theta}{r^2}$. Also determine the streamlines. Show that the motion is irrotational. **06**

*****The End*****

Khulna University, Khulna
Mathematics Discipline
4th Year, Term-I, Examination Session: 2023-2024
Course No: Math-4105
Full Title of Course: Hydrodynamics
Full Marks: 60, Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

1. (a) Define viscosity with geometrical interpretation. State and prove Bernoulli's theorem for incompressible fluid. 05
 (b) Discuss the conservation of mass in hydrodynamic point of view. Derive the equation of continuity for incompressible and irrotational flow. 05
2. (a) Prove that the stream function is constant along streamline. 05
 (b) Show that the velocity potential $\phi = \frac{1}{2}a(x^2 + y^2 - 2z^2)$ satisfies the Laplacian equation. 05
 Also determine the streamlines.
3. (a) Define complex potential and velocity of a fluid motion. State and prove the circle theorem for complex potential. 05
 (b) Show that $u = \frac{-2xyz}{(x^2 + y^2)^2}$, $v = \frac{(x^2 - y^2)z}{(x^2 + y^2)^2}$, $w = \frac{y}{x^2 + y^2}$ are the velocity components of the possible fluid motion. Is this motion irrotational? 05
4. (a) What is boundary surface? Show that the variable ellipsoid $\frac{x^2}{a^2k^2t^4} + kt^2 \left\{ \left(\frac{y}{b} \right)^2 + \left(\frac{z}{c} \right)^2 \right\} = 1$ is a possible form for the boundary surface of a liquid at any time t . 05
 (b) Show that the velocity potential $\phi = \frac{1}{2} \log \left(\frac{(x+a)^2 + y^2}{(x-a)^2 + y^2} \right)$, gives a possible motion, determines the streamlines, and shows that the curves of equal speed are the ovals of Cassini given by $rr' = \text{constant}$. 05

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

5. (a) Two sources of strength m (each) are placed at the points $(1, 0)$ and $(-1, 0)$ and sinks of equal strength m are placed at $(0, 1)$ and $(0, -1)$. Show that the streamlines through these four points is a circle. 05
 (b) Show that, $u = 2cxy$, $v = c(a^2 + x^2 - y^2)$ are the velocity component of a possible fluid motion. Determine the stream function and sketch them. 05
6. (a) A stream in a horizontal pipe, after passing a contraction in the pipe at which its sectional area is A , is delivered at atmospheric pressure at a place where the sectional area is B . Show that if a side tube connected with the pipe at the former place, water will be sucked up through it into the pipe from a reservoir a depth $\frac{s^2}{2g} \left(\frac{1}{A^2} - \frac{1}{B^2} \right)$ below the pipe; where s is the delivery per second. 05
 (b) Find the ϕ and ψ from $w = Az^n$. Also discuss the cases if i) $n > 1$ ii) $1/2 < n < 1$ and (iii) $n = 1/2$. 05
7. (a) State and prove the Blasius theorem. Give a comparative discussion on Blasius theorem and circle theorem. 06
 (b) Prove that a flow is irrotational for the given velocity $\underline{q} = \frac{-yi + xj}{x^2 + y^2}$. Calculate the circulation round (i) a square with corners $(1, 0)$, $(2, 0)$, $(2, 1)$ and $(1, 1)$ and (ii) a unit circle with center at the origin. 04
8. (a) State and prove Kelvin's minimum energy theorem. 05
 (b) Find the stream function of two-dimensional motion due to two equal sources and an equal sink midway between them. Sketch the streamlines. 05

***** THE END *****

Khulna University
 Science, Engineering and Technology School
 Discipline: **Mathematics**, Program: **B.Sc. (Hons.)**
 Year: **Fourth** Term: **I** Examination-2026; Session: **2024-2025**
 Course Code: **Math-4105**
 Course Title: **Hydrodynamics**



Total Marks: **60**, Time: **03** Hours, Credit: **03**

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | Marks |
|---|--------------|
| 1. (a) Define streamline and path line. State and prove Bernoulli's theorem for compressible fluid. | 05 |
| (b) Derive the continuity equation for incompressible and compressible fluids in vector form | 05 |
| 2. (a) Define complex potential. Find the complex potential for the streaming motion flow past an elliptic cylinder, the stream U making an angle α with the real axis. | 05 |
| (b) Show that the velocity potential $\phi = \frac{1}{2}a(x^2 + y^2 - 2z^2)$ satisfies Laplace's equation. Also determine the streamlines. | 05 |
| 3. (a) Established the equation of motion for inviscid fluid in the form;
$\frac{dq}{dt} - q \wedge (\nabla \wedge q) = F - \frac{1}{\rho} \nabla p - \frac{1}{2} \nabla q^2.$ | 05 |
| (b) Show that $u = \frac{-2xyz}{(x^2 + y^2)^2}$, $v = \frac{(x^2 - y^2)z}{(x^2 + y^2)^2}$, $w = \frac{y}{x^2 + y^2}$ are the velocity components of the possible fluid motion. Is this motion irrotational? | 05 |
| 4. (a) Derive the acceleration of a fluid particle in a Cartesian coordinate system. | 05 |
| (b) Prove that the velocity function $\psi = U(r + \frac{a^2}{r}) \cos \theta$ represents a streaming motion past a fixed circular cylinder. | 05 |

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | |
|--|-----------|
| 5. (a) Define source and sink with geometrical configuration. Find the complex potential for equal source and sink in a stream. | 05 |
| (b) Two sources of strength m are placed at the points $(a,0)$ and $(-a,0)$ and the sink of strength $2m$ is placed at the origin, show that $(x^2 + y^2)^2 = a^2(x^2 - y^2 + 2\lambda xy)$ where λ is the variable parameter. | 05 |
| 6. (a) Define vortex line. Find the necessary and sufficient condition that vortex lines may be at right angles to streamlines. | 05 |
| (b) A vortex pair is situated within a cylinder. Show that it will remain at rest if the distance of either from the center is given by $(\sqrt{5} - 2)^{1/2}a$, where a is the radius of the cylinder. | 05 |
| 7. (a) State and prove circle theorem. | 05 |
| (b) Find the center of pressure of a triangular area, the depth of whose vertices are α, β, γ from the free surface. | 05 |
| 8. (a) Find the center of pressure of a triangular area whose one vertex is on the free surface. | 05 |
| (b) Define stagnation point. Prove that a streamline is divided into two branches at the point of stagnation and perpendicular to each other when the motion is irrotational. | 05 |

***** THE END *****

Khulna University
Science, Engineering and Technology School
Discipline: **Mathematics**, Program: **B.Sc. (Hons.)**
Year: **Fourth**, Term: **I** Examination-2026; Session: **2024-2025**
Course Code: **0541 12 Math 4101**
Course Title: **Hydrodynamics and Hydrostatics**
Total Marks: **60**, Time: **03** Hours, Credit: **03**



- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

1. (a) Define stream tube, stream filament, rotational and irrotational flow. Distinguish between stream lines and path lines. [L4; CLO1] **05**
(b) Derive a relation between local and individual rates of changes. Also briefly discuss about velocity potential or velocity function. [L5; CLO1] **05**
2. (a) Given the velocity components: $u = \frac{ax - by}{x^2 + y^2}, v = \frac{ay + bx}{x^2 + y^2}, w = 0$. Explain these components for a possible fluid flow. Also test its rotationality. [L4; CLO2] **04**
(b) Deduce the equation of conservation of mass of fluid. If $q = -i(3y^2) - j(6x)$, determine the stream lines at (1, 1). [L5; CLO1] **06**
3. (a) State and prove Bernoulli's theorem for steady motion of an inviscid fluid. [L5; CLO2] **05**
(b) Derive Euler's equation of motion in the form $\frac{\partial q}{\partial t} + \frac{1}{2} \nabla q^2 - q \times \text{curl} q = F - \frac{1}{\rho} \nabla p$. [L5; CLO2] **05**
4. (a) Find the expression to the complex potential for the streaming motion past an ellipse in terms of the elliptic co-ordinates and hence find the stream function. [L6; CLO3] **05**
(b) Prove that the velocity function $\psi = U(r + \frac{a^2}{r}) \cos \theta$ represents a streaming motion past a fixed circular cylinder. [L6; CLO3] **05**

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

5. (a) Define source and sink. What are the values of velocity, stream function and velocity potential at the sink and source. Find the stream function if a source is in a uniform stream. [L1; CLO3] **05**
- (b) What is doublet? Derive the complex potential due to a doublet. [L5; CLO2,3] **05**
6. (a) State and prove Blasius theorem. [L5; CLO3] **05**
- (b) Discuss the motion represented by $W = \frac{1}{2}U \frac{a^3}{z^2}$, and show that the stream lines are lemniscates. [L5; CLO3] **05**
7. (a) Find the strength of a circular vortex. Show that the circulation can be considered as the measure of velocity potential around a closed circuit. [L2; CLO4] **05**
- (b) Calculate the circulation of the velocity field $\vec{q} = \frac{-yi + xj}{x^2 + y^2}$ round a square with corners (1, 0), (2, 0), (2, 1) and (1, 1). [L4; CLO4] **05**
8. (a) Find the center of pressure of a triangular area, the depth of whose vertices are α, β, γ from the free surface. [L5; CLO4] **05**
- (b) A regular hexagon is immersed in a liquid with one side in the surface. Show that the depth of its centre of pressure is to that of its centre of gravity as 23:18. [L6; CLO4] **05**

***** THE END *****